

2(a)	<i>idea of</i> no (clear), relationship / trend ; or as latitude increases, (bird) species diversity / numbers of (bird) species, (slightly) decreases / (might) show a downward trend ;	1
2(b)(i)	any three from: 1 data (for latitude and number of species) is, paired / linked ; 2 number of bird species and latitude / (i.e. the data points) within samples are independent of each other ; 3 the <u>data</u>, is ordinal / is interval / is discrete / can be ranked ; 4 (scatter) graph / Fig. 2.2, suggests a, (decreasing) relationship / correlation ; 5 there are, 17 (paired) observations / more than 5 (paired) observations ; 6 <i>idea that</i> bird species selected randomly / AW ;	3
2(b)(ii)	<i>idea that</i> there is no, correlation / relationship, between latitude and the number of (bird) species ;	1
2(b)(iii)	any four from: 1 critical value (for $p = 0.05\%$) = 0.485 or - critical value for $p = 0.01\%$ = 0.615 ; 2 value for r_s / calculated value, is < than critical value ; 3 null hypothesis is accepted ; 4 there is no <u>significant</u> correlation (between latitude and number of bird species) ; 5 there is a (weak) negative correlation (between latitude and number of bird species as shown by the negative value for r_s) ;	4

Question	Answer	Marks
2(c)	any three from all marking points are ideas that: <i>limitations of the data</i> 1 only used counts of bird species (may not apply to all species) ; 2 do not know the, number of individuals of each species / relative abundance of each species / species evenness ; <i>limitations of sampling</i> <i>birds</i> 3 hard to identify some (bird) species / miss some (bird) species / some (bird) species look similar / miss birds that fly away / birds migrate / only resident birds or species counted / qualified ref to volunteers / AW ; <i>latitudes sampled</i> 4 nothing above 40N / nothing before 37N / need wider range of latitudes / AW ; 5 no data on latitudes south of the equator ; <i>sampling sites</i> 6 not random / only sampled in certain locations / sample locations do not cover the whole of the peninsula / many near the coast / locations may not be representative of the peninsula / only 17 sites / sampling sites do not extend beyond the peninsular / AW ; <i>habitats</i> 7 no information on habitats (at locations) / not all location may be equally suited for birds / different habitats have different conditions / AW ; <i>timings</i> 8 only sampled, on one day / in one year / sampled in, winter / one season ; ora	3

Question	Answer	Marks
2(d)	any three from: 1 catch birds within area and count total captured and mark and release them ; 2 detail e.g. <i>idea of</i>: marking technique not, harmful / too obvious e.g. <i>idea of</i>: giving time for populations to mix before second sample taken – days or weeks not more ; 3 catch second sample and count total number of birds and number of marked birds ; 4 <i>ref. to</i> Lincoln / Petersen, index ;	3